

Level 5 – final integration – practical assignment

a)

In both gsr-dc and store:

In root, install and enable sudo

```
$ apt-get install sudo
```

```
$ sudo nano /etc/sudoers
```

```
add: rafaelswr ALL=(ALL:ALL)ALL
```

```
$ sudo apt-get update
```

Conf network gsr-dc debian for store

```
auto ens33
```

```
inet ens33 iface static
```

```
address 192.168.220.146
```

```
netmask 255.255.255.0
```

```
gateway 192.168.220.2
```

Mudar nome da maquina:

```
$ sudo hostnamectl set-hostname gsr-dc
```

Em /etc/hosts:

```
192.168.220.146 gsr-dc.gsr.ipb.local gsr-dc
```

ver dominio:

```
$ hostname
```

```
$ hostname -i
```

```
$hostname -f
```

Conf network gsr-dc debian for account

-> static ip

Sudo nano /etc/network/interfaces

auto ens33

inet ens33 iface static

address 192.168.220.146

netmask 255.255.255.0

gateway 192.168.220.2

Mudar nome da maquina:

\$ sudo hostnamectl set-hostname store

Em /etc/hosts:

192.168.220.145 store.gsr.ipb.local store

ver dominio:

\$ hostname

\$ hostname -i

\$hostname -f

->set up samba

sudo apt-get install samba winbind

sudo systemctl stop smbd winbind nmbd

sudo systemctl disable smbd

sudo systemctl unmask samba-ad-dc.service

sudo systemctl enable samba-ad-dc.service

-> rename smb.conf file, because set up samba ad-dc will create a new smb.conf

```
Cd /etc/samba/smb.conf  
sudo mv smb.conf smb.conf.old
```

Interactive mode:

```
$ sudo samba-tool domain provision --use-rfc2307 --interactive  
Realm: gsr.ipb.local  
domain: gsr  
server Role: dc  
Dns backend : SAMBA_INTERNAL  
DNS Forwarder IP : 192.168.220.2
```

```
Sudo systemctl restart samba-ad-dc.service  
sudo systemctl status samba-ad-dc.service
```

Add record of hosts of network

```
Sudo samba-tool dns add gsr-dc gsr.ipb.local store A  
192.168.220.145 -UAdministrator%Rafael.1999
```

See all records, samba internal dns

```
sudo samba-tool dns query gsr-dc gsr.ipb.local @ ALL -  
UAdministrator%Rafael.1999
```

Criar o utilizador para o dominio, é automaticamente adicionado ao grupo Domain Users:

```
Samba-tool user create a39041 --given-name="José" --surname="Borges" password=Rafael.1999
```

Ver utilizadores do dominio
sudo wbinfo -u

Ver Grupo do dominio:

Sudo samba-tool group list

Ver membros do grupo Domain Users:

Sudo samba-tool group listmembers "Domain Users"

```
#####sudo samba-tool domain join gsr.ipb.local DC -UAdministrator%Rafael.1999#####
```

Create directorio para partilha

sudo mkdir /mnt/gsr

sudo chmod 775 /mnt/gsr

Samba-share

[gsr]

Comment = GSR share

valid users = @"Domain Users"

path = /mnt/gsr

read only = no

browsable = yes

writable = yes

public = no

\$ sudo systemctl restart samba-ad-dc.service

Install snmp

Sudo apt-get install snmpd snmp-mibs-downloader mrtg snmp

Edit snmpd.conf file

Sudo nano /etc/snmp/snmpd.conf

Add agentAddress udp:192.168.220.146:161

sysLocation "Location"

sysContact "Contact Information"

Sudo systemctl restart snmpd

Test snmp

```
Sudo snmpwalk -v 1 -c public 192.168.50.129
Sudo snmpwalk -v 1 -c public 192.168.220.146 1.3.6.1.4.1.2021.11.9.0
```

Configure mrtg and apache2

```
Sudo apt-get install apache2
```

```
Sudo mkdir /var/www/mrtg
```

Permissions:

```
sudo chown -R $USER:$USER /var/www/mrtg
sudo chmod -R 755 /var/www/mrtg
```

```
Generate virtualHostFile ( /etc/apache2/sites-available/mrtg.conf
```

```
GNU nano 5.4                                mrtg.conf
<VirtualHost *:80>

    DocumentRoot /var/www/mrtg
    ServerName 192.168.220.146

    <Directory /var/www/mrtg>
        Options Indexes FollowSymLinks MultiViews
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
```

Back to apache, enable virtual host, on /etc/apache2/sites-available

```
sudo a2ensite mrtg
disable the default Apache configurations
sudo a2dissite 000-default.conf
Sudo systemctl restart apache2.service
```

Config mrtg:

```
sudo cfgmaker public@192.168.50.129 --global 'Options[_]: bits,growright' --  
output /etc/mrtg.cfg
```

Edit mrtg.cfg, add:

RunAsDaemon: yes

LoadMIBs: /usr/share/snmp/mibs/UCD-SNMP-MIB.txt

Options[_]: growright, bits

Target[cpu]:

1.3.6.1.4.1.2021.11.9.0&1.3.6.1.4.1.2021.11.10.0:public@192.168.50.129

MaxBytes[cpu]: 100

Title[cpu]: CPU Usage

PageTop[cpu]: <h1>Cpu Space</h1>

YLegend[cpu]: % of CPU Utilization

ShortLegend[cpu]: %

LegendI[cpu]: CPU Utilization:

LegendO[cpu]:

Options[cpu]: gauge, nopercent

Target[disk]:

1.3.6.1.4.1.2021.9.1.9.1&1.3.6.1.4.1.2021.9.1.7.1:public@192.168.50.129

MaxBytes[disk]: 100000000

Title[disk]: Disk Space

PageTop[disk]: <h1>Disk Space</h1>

YLegend[disk]: Disk Space (MB)

ShortLegend[disk]: MB

LegendI[disk]: Disk Space in:

[40 linhas lidas]

Generate index:

```
sudo indexmaker /etc/mrtg.cfg --output /var/www/mrtg/index.html
```

```
sudo env LANG=C /usr/bin/mrtg /etc/mrtg.cfg
```

Test

Store.gsr.ipb.local

A RAID6 array supported by set of disks to store files (any size).

Adicionados 6 discos, 4 ativos e 2 de hotspare de tamanho de 200MB.

```
$sudo lsblk -o NAME,SIZE
```

Criar o array de raid 6:

```
$ sudo mdadm -create -verbose /dev/md0 --level=6 -raiddevices=4 /dev/sda  
/dev/sdc /dev/sdd /dev/sde --spare-devices=2 /dev/sdf /dev/sdg
```

Criar physical volume sobre o raid 6, combinado os discos como um so disco fisico, usado para guardar dados

```
$sudo apt-get install lvm2
```

```
$ sudo pvcreate /dev/md0
```

```
$ sudo pvs
```

```
$sudo vgcreate /dev/md0 vg0
```

```
$sudo vgs
```

```
$sudo lvcreate -L 200M vg0 -n lv0
```

Create filesystem in JFS to store files:

```
$ sudo apt install jfsutils
```

Criar diretorio

```
$sudo mkdir /mnt/gsr
```

```
$sudo mkfs.jfs /dev/vg0/lv0
```

In \$ sudo blkid, copy UUID da particao virtual criada e colar no /etc/fstab:

```
UUID “...” /mnt/gsr jfs defaults 0 0
```

```
$ sudo mount -a
```

user snmpv3 with auth and privacy

```
$ sudo apt-get install snmp, snmpd
```

Add contrib non-free to /etc/apt/sources.list

```
$ sudo apt-get install snmp-mibs-downloader libsnmp-dev
```

```
$ sudo apt-get update
```

```
$ sudo systemctl stop snmpd
```

Create a user snmpv3:

```
$sudo net-snmp-config --create-snmpv3-user (-ro) -a SHA -A  
"rafaelswr1223" -x AES -X "rafaelswr1223priv" rafaelswr
```

->In the /etc/snmp/snmpd.conf

Change sysLocation and sysContact, like:

```
SysLocation "Location"  
sysContact "Me"
```

Change AgentAddress, to IP address or hostname of the device that is running the SNMP service:

```
AgentAddress udp:192.168.220.145:161  
#agentAddress udp:161,udp6: [::1]:161
```

Defining a read-write SNMPv3 user named **rafaelswr** with privacy (also known as encryption) enabled.

```
rwuser rafaelswr priv
```

Para conferir, podemos ir ao ficheiro /var/lib/snmp/snmpd.conf, antes de fazer restart do snmp

```
$ sudo nano /var/lib/snmp/snmp.conf
```

Aparece a seguinte linha:

```
CreateUser rafaelswr SHA "rafaelswr1223" AES  
"rafaelswr1223priv"
```

```
$sudo systemctl restart snmpd.service
```

```
$sudo apt-get update
```

```
$sudo systemctl status snmpd.service
```


Test user:

```
$ snmpwalk -v 3 -u rafaelswr -l authPriv -a SHA -A  
"rafaelswr1223" -x AES -X "rafaelswr1223priv"  
192.168.220.145
```

Instalar DHCP and Permit ddns with samba-ad-dc internal backend dns server

In the dhcp server:

```
$ sudo apt-get install isc-dhcp-server
```

```
$ sudo nano /etc/dhcp/dhcpd.conf
```

```
option domain-name "gsr.ipb.local";
```

```
option domain-name-servers 192.168.220.146;
```

```
ddns-updates on;
```

```
ddns-update-style interim;
```

```
ignore client-updates;
```

```
update-static-leases on;
```

```
authoritative;
```

```
zone gsr.ipb.local. {
```

```
    #primary 192.168.50.129;
```

```
    primary gsr-dc.gsr.ipb.local;
```

```
}
```

```
Zone 50.168.192.in-addr.arpa {
```

```
#primary 192.168.50.129;
```

```
    primary gsr-dc.gsr.ipb.local;
```

```
}
```

```
$sudo systemctl restart isc-dhcp-server.service
```

Test MRTG:

```
##Disk Statitics##
```